

Regular-variation limits for nonmultiplicative divisibility Gram matrices

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Abstract

Let $\sigma < 1/2$, write $\rho = 1 - 2\sigma$, and let L be a positive measurable slowly varying function, bounded above and away from zero on compact subintervals of $[1, \infty)$. We consider the divisibility matrices $T_N(r, m) = (r/m)^{-\sigma} L(r/m)$ when $m \mid r \leq N$, with all other entries zero. Without assuming multiplicativity, we prove that the normalized Gram operators $\rho T_N^* T_N / (N^\rho L(N)^2)$ converge to the classical LCM operator with entries $(mn)^\sigma / \text{lcm}(m, n)$ in every Schatten ideal $\mathcal{S}_q$ such that $q\rho > 1$, including $0 < q < 1$. The range is exact: for $q\rho \leq 1$, the difference does not belong to $\mathcal{S}_q$ for any finite N . A positive-congruence argument gives eigenvalue bounds $C_\eta j^{-\eta}$, uniform in N , for every $0 < \eta < \rho$. These bounds turn an explicit regular-variation limit of the Gram entries into operator-norm and then ideal convergence. The sharpness uses the known eigenvalue asymptotics of the LCM operator. We also obtain cumulative-normalization, fixed-index singular-value, and spectral-moment consequences, allowing nonconstant oscillatory slowly varying factors.

Keywords: divisibility matrix; regular variation; LCM kernel; Schatten ideal; singular-value asymptotics.

1 Introduction

Divisibility matrices replace the difference of two indices in an ordinary Toeplitz matrix by their integer ratio. Their Gram matrices therefore couple analytic growth of coefficients with the arithmetic relation of having a common multiple. For power coefficients, the limiting kernel is expressed by the least common multiple. The question studied here is whether that same kernel persists when the coefficients are modified by an arbitrary slowly varying factor, without a multiplicative assumption, and in which operator ideals the limit holds.

Fix $\sigma < 1/2$ and set $\rho = 1 - 2\sigma > 0$. Throughout the paper, $L : [1, \infty) \rightarrow (0, \infty)$ is measurable and slowly varying, and both L and $1/L$ are bounded on every compact subinterval of $[1, \infty)$. With $a(k) = k^{-\sigma} L(k)$, define on $\ell^2(\mathbb{N})$

$$T_N(r, m) = \begin{cases} a(r/m), & m \mid r, \quad 1 \leq m \leq r \leq N, \\ 0, & \text{otherwise,} \end{cases} \quad A_N = \frac{\rho}{N^\rho L(N)^2} T_N^* T_N. \quad (1.1)$$

Thus T_N and A_N are finite matrices extended by zero to the same infinite-dimensional Hilbert space. Let E_σ denote the classical LCM operator

$$E_\sigma(m, n) = \frac{(mn)^\sigma}{\text{lcm}(m, n)}, \quad m, n \geq 1. \quad (1.2)$$

Its compactness, positivity, and eigenvalue asymptotics are known; the precise input is recalled in Theorem 2.1.

For $q > 0$, membership in $\mathcal{S}_q$ means summability of the q th powers of the singular values. We use the standard quasi-norm $\|K\|_{\mathcal{S}_q} = (\sum_j s_j(K)^q)^{1/q}$; it is a norm when $q \geq 1$. All eigenvalue

Table 1: The coefficient hypotheses and convergence scopes being compared. The preprint-version qualification in the second row is essential.

Source	Coefficient setting	Relevant conclusion
Hilberdink (2017)	Complete multiplicativity, or multiplicativity with extra hypotheses; regularly varying cumulative energy	Hilbert–Schmidt and trace-class results under stated conditions
Hilberdink–Pushnitski, 2021 version	Pure powers $k^{-\sigma}$	LCM spectral asymptotics; Gram convergence for even q , $q\rho > 1$
Theorem 1.1	$k^{-\sigma}L(k)$ with arbitrary admissible slowly varying L ; no multiplicativity	Gram convergence for every real $q > 0$ with $q\rho > 1$, and exact failure outside this range

sequences below are ordered nonincreasingly, with multiplicity, and finite sequences are continued by zeros.

Theorem 1.1 (Regular-variation Gram limit). *Under the hypotheses above, the following statements hold.*

1. For every $0 < \eta < \rho$, there is a constant $C_\eta = C(\sigma, L, \eta)$, independent of N and j , such that

$$\lambda_j(A_N) \leq C_\eta j^{-\eta}, \quad \lambda_j(E_\sigma) \leq C_\eta j^{-\eta} \quad (N, j \geq 1). \quad (1.3)$$

2. $\|A_N - E_\sigma\| \rightarrow 0$ as $N \rightarrow \infty$.

3. For every real $q > 0$ with $q\rho > 1$,

$$\|A_N - E_\sigma\|_{\mathcal{S}_q} \rightarrow 0. \quad (1.4)$$

If $q\rho \leq 1$, then $A_N - E_\sigma \notin \mathcal{S}_q$ for every N .

The coefficient condition is pointwise regular variation, not multiplicativity. In particular, it admits $L(x) = (1 + \log x)^\beta$ for every real β , and also positive oscillatory slowly varying functions without a limit. The endpoint $\eta = \rho$ is not asserted in the uniform bound (1.3); an arbitrarily small loss suffices to obtain every ideal in the exact range $q\rho > 1$.

1.1 Antecedents and ownership of the inputs

Hilberdink developed regular-variation limits for multiplicative Toeplitz matrices [2]. For completely multiplicative coefficients, his Theorem 2.2 and Corollary 2.3 give Hilbert–Schmidt and trace-norm convergence under the corresponding growth conditions. Theorems 3.1 and 3.2 treat multiplicative coefficients with additional local-series and summability hypotheses. Regular variation of the cumulative coefficient energy is a central hypothesis there. Our coefficients need not be multiplicative, but we require the specific pointwise form $k^{-\sigma}L(k)$; neither statement should be read as containing the other in full generality.

Hilberdink and Pushnitski studied the LCM family that contains (1.2), proving its spectral asymptotics and relating it to power-coefficient multiplicative Toeplitz matrices [3]. Theorem 2.1 in the accessed 2021 preprint version, arXiv:2110.14323v1, states the power-coefficient Gram convergence for even integers q satisfying $q\rho > 1$. When $L = 1$, Theorem 1.1 gives every real q in that range. This is a comparison with that explicitly identified version, not a claim that the parity restriction remains an open problem in subsequent literature. The known LCM eigenvalue asymptotic and its arithmetic factorization remain attributed to Hilberdink and Pushnitski.

The regular-variation tools are the classical uniform convergence theorem and Potter bounds [1]; the compact-operator tools are also standard [4]. The argument specific to the present family combines uniform domination by a shifted LCM kernel with a positive diagonal congruence. This gives an N -independent summable spectral tail after choosing an exponent strictly inside the allowed range. Counting-measure dominated convergence then works for $0 < q < 1$ as well as for Banach ideals.

Sections 2 and 3 set up the classical inputs and prove the entrywise limit, including its small-index tail. Section 4 obtains uniform spectral control and operator-norm convergence. Section 5 proves the full ideal range and sharpness. Section 6 gives the normalization and singular-value consequences.

2 Framework and classical inputs

2.1 Regular variation

A positive measurable function L is slowly varying at infinity if

$$\frac{L(tx)}{L(x)} \rightarrow 1 \quad (x \rightarrow \infty) \quad (2.1)$$

for each fixed $t > 0$. The measurability and local upper and lower bounds in our standing assumptions are retained throughout. The classical uniform convergence theorem gives, for every $0 < c < C < \infty$,

$$\sup_{t \in [c, C]} \left| \frac{L(tN)}{L(N)} - 1 \right| \rightarrow 0. \quad (2.2)$$

Only values with $tN \geq 1$ will be used. Potter's estimate, together with the local bounds, gives for every $\varepsilon > 0$ a constant K_ε such that

$$\frac{L(x)}{L(y)} \leq K_\varepsilon \max \left\{ \left(\frac{x}{y} \right)^\varepsilon, \left(\frac{x}{y} \right)^{-\varepsilon} \right\} \quad (x, y \geq 1). \quad (2.3)$$

These are standard regular-variation facts [1]; the global form (2.3) under local upper and lower bounds is also stated in Section 1.1 of [2].

For clarity, the extension of eventual Potter bounds to all $x, y \geq 1$ does not discard any behavior on a compact interval. Choose $b \geq 1$ so that the eventual estimate holds when both arguments are at least b . If $x \geq b > y$, factor the ratio as $[L(x)/L(b)][L(b)/L(y)]$ and use the positive minimum bound on $[1, b]$; then $(x/b)^\varepsilon \leq (x/y)^\varepsilon$. If $x < b \leq y$, use the analogous factorization and the upper bound on $[1, b]$. If both arguments lie in $[1, b]$, their ratio is bounded by the local upper-to-lower bound, while the maximum on the right of (2.3) is at least one. Enlarging the constant proves the displayed global estimate.

2.2 The classical LCM operator

We use the following specialization of an existing spectral theorem. It is an input, not a result proved anew here.

Theorem 2.1 (Hilberdink–Pushnitski). *For every real $s < 1/2$, the matrix*

$$E_s(m, n) = \frac{(mn)^s}{\text{lcm}(m, n)} \quad (2.4)$$

defines a positive, compact, injective operator on $\ell^2(\mathbb{N})$. There is a constant $\kappa(s, 1) > 0$ such that

$$\lambda_j(E_s) \sim \kappa(s, 1) j^{-(1-2s)} \quad (j \rightarrow \infty). \quad (2.5)$$

Consequently, for every real $q > 0$,

$$E_s \in \mathcal{S}_q \iff q(1 - 2s) > 1. \quad (2.6)$$

This is Theorem 1.1 of [3], specialized to the second kernel parameter equal to one. The theorem numbering and the specialization were checked in arXiv:2110.14323v1 (2021); the bibliography also identifies the subsequent journal publication. The equivalence (2.6) follows immediately from (2.5) and the positive constant there. No new value or formula for $\kappa(s, 1)$ is claimed.

2.3 Singular values for all positive ideal exponents

For a compact operator K , write $s_1(K) \geq s_2(K) \geq \dots \geq 0$ for its singular values. We recall the standard inequality below and include its short proof because it is the step used in the quasi-Banach range. Approximation numbers on a Hilbert space satisfy

$$s_j(K) = \inf_{\text{rank } F < j} \|K - F\|. \quad (2.7)$$

This follows from truncating the singular-value decomposition and the min-max principle; see, for example, [4] for the compact-operator framework.

Lemma 2.2 (Singular-value sum bound). *For compact U, V and integers $j, k \geq 1$,*

$$s_{j+k-1}(U + V) \leq s_j(U) + s_k(V). \quad (2.8)$$

For every $q > 0$, the class $\mathcal{S}_q$ is closed under sums and differences.

Proof. Choose finite-rank approximants F, G with ranks less than j, k , respectively. Then $\text{rank}(F + G) \leq j + k - 2$, and

$$s_{j+k-1}(U + V) \leq \|U - F\| + \|V - G\|.$$

Taking the two infima proves (2.8). For closure under sums, take $k = j$, use monotonicity for the intervening even index, and apply

$$(x + y)^q \leq c_q(x^q + y^q), \quad c_q = \max\{1, 2^{q-1}\}, \quad x, y \geq 0.$$

The resulting summable majorant proves the assertion for all $q > 0$. Negation does not change singular values, so differences are included. $\square$

For positive compact operators, singular values and eigenvalues agree. We also use the standard min-max implications $0 \leq U \leq V \Rightarrow \lambda_j(U) \leq \lambda_j(V)$ and

$$|\lambda_j(U) - \lambda_j(V)| \leq \|U - V\| \quad (2.9)$$

for positive compact U, V . All operator inequalities here are in quadratic-form order. Entrywise domination of nonnegative matrices is not, by itself, such an inequality.

3 Gram entries and their regular-variation limit

3.1 The exact common-multiple identity

Put $\ell = \text{lcm}(m, n)$. Multiplying the matrices in (1.1) gives

$$\begin{aligned} A_N(m, n) &= \frac{\rho}{N^\rho L(N)^2} \sum_{\substack{r \leq N \\ m|r, n|r}} (r/m)^{-\sigma} (r/n)^{-\sigma} L(r/m) L(r/n) \\ &= \frac{\rho(mn)^\sigma}{N^\rho L(N)^2} \sum_{k \leq N/\ell} (k\ell)^{-2\sigma} L(k\ell/m) L(k\ell/n). \end{aligned} \quad (3.1)$$

The sum is empty when $\ell > N$, so the identity also covers an index greater than N . This elementary Gram identity is the common-multiple formula behind the multiplicative Toeplitz literature; compare Proposition 2.1 of [2]. Here it uses no multiplicativity. Every entry of A_N is nonnegative, and A_N is positive semidefinite because it is a positive scalar multiple of a Gram matrix.

Lemma 3.1 (Power-sum bound). *For every real $\alpha < 1$, there is $C_\alpha < \infty$ such that*

$$\sum_{1 \leq k \leq X} k^{-\alpha} \leq C_\alpha X^{1-\alpha} \quad (X \geq 1). \quad (3.2)$$

Proof. If $\alpha < 0$, each summand is at most $X^{-\alpha}$ and there are at most X terms. If $0 \leq \alpha < 1$, integral comparison gives $\sum_{k \leq X} k^{-\alpha} \leq 1 + \int_1^X t^{-\alpha} dt$, which is bounded by a constant times $X^{1-\alpha}$. $\square$

3.2 Uniform entrywise domination

Proposition 3.2. *For every $\varepsilon > 0$ with $2\varepsilon < \rho$, there is a constant $C = C(\sigma, L, \varepsilon)$ such that*

$$0 \leq A_N(m, n) \leq C E_{\sigma+\varepsilon}(m, n) \quad (N, m, n \geq 1). \quad (3.3)$$

Proof. The assertion is immediate for an empty Gram sum. Otherwise write $r = k\ell \leq N$. Its arguments satisfy $1 \leq r/m, r/n \leq N$. The global Potter estimate (2.3) therefore gives

$$\frac{L(r/m)}{L(N)} \leq K_\varepsilon (mN/r)^\varepsilon, \quad \frac{L(r/n)}{L(N)} \leq K_\varepsilon (nN/r)^\varepsilon. \quad (3.4)$$

Let $\alpha = 2\sigma + 2\varepsilon < 1$. Substituting these bounds in (3.1) and using Lemma 3.1 yields

$$\begin{aligned} A_N(m, n) &\leq C(mn)^{\sigma+\varepsilon} N^{-\rho+2\varepsilon} \ell^{-\alpha} \sum_{k \leq N/\ell} k^{-\alpha} \\ &\leq C'(mn)^{\sigma+\varepsilon} N^{-\rho+2\varepsilon} \ell^{-\alpha} (N/\ell)^{1-\alpha} = C' \frac{(mn)^{\sigma+\varepsilon}}{\ell}. \end{aligned}$$

The powers of N cancel because $1 - \alpha = \rho - 2\varepsilon$. This proves the proposition. $\square$

The estimate (3.3) is only entrywise domination. We will use it to bound a bilinear form after taking coordinatewise absolute values. Positivity and a separate congruence argument are needed to obtain operator-order and eigenvalue bounds.

3.3 Entrywise convergence with the small-index tail controlled

Proposition 3.3. *For every fixed pair $m, n \geq 1$,*

$$A_N(m, n) \longrightarrow E_\sigma(m, n). \quad (3.5)$$

Moreover, with $F(N) = \sum_{k \leq N} k^{-2\sigma} L(k)^2$,

$$F(N) \sim \frac{N^\rho L(N)^2}{\rho}. \quad (3.6)$$

Proof. Fix m, n , put $\ell = \text{lcm}(m, n)$, and write $h = \ell/N$ and $t_k = kh$. Formula (3.1) becomes

$$A_N(m, n) = \frac{\rho(mn)^\sigma}{\ell} h \sum_{k \leq 1/h} t_k^{-2\sigma} \frac{L(Nt_k/m)}{L(N)} \frac{L(Nt_k/n)}{L(N)}. \quad (3.7)$$

All arguments of L in this sum are at least one, because $Nt_k/m = k\ell/m \geq 1$, and similarly for n .

Fix $0 < \delta < 1$. On the portion $\delta \leq t_k \leq 1$, the two ratios in (3.7) converge uniformly to one by (2.2), applied on $[\delta/m, 1/m]$ and $[\delta/n, 1/n]$. The function $t^{-2\sigma}$ is continuous and bounded on $[\delta, 1]$. Consequently, this portion of the sum, including its factor h , tends to

$$\int_{\delta}^1 t^{-2\sigma} dt. \quad (3.8)$$

It remains to justify removal of the lower cutoff. Choose $\varepsilon > 0$ with $2\varepsilon < \rho$, and again set $\alpha = 2\sigma + 2\varepsilon < 1$. By (3.4), the product of ratios in (3.7) is at most $C(mn)^{\varepsilon} t_k^{-2\varepsilon}$. If $\delta/h < 1$, there are no terms with $t_k < \delta$. Otherwise Lemma 3.1 gives

$$\begin{aligned} h \sum_{t_k < \delta} t_k^{-2\sigma} \frac{L(Nt_k/m)}{L(N)} \frac{L(Nt_k/n)}{L(N)} &\leq C_{m,n} h \sum_{k < \delta/h} (kh)^{-\alpha} \\ &\leq C'_{m,n} \delta^{1-\alpha}. \end{aligned} \quad (3.9)$$

The last bound is uniform in N and tends to zero as $\delta \downarrow 0$. The omitted integral $\int_0^{\delta} t^{-2\sigma} dt$ also tends to zero, since $2\sigma < 1$. Letting first $N \rightarrow \infty$ and then $\delta \downarrow 0$ in (3.7) proves

$$A_N(m, n) \rightarrow \frac{\rho(mn)^{\sigma}}{\ell} \int_0^1 t^{-2\sigma} dt = \frac{(mn)^{\sigma}}{\ell}.$$

Finally $A_N(1, 1) = \rho F(N)/(N^{\rho} L(N)^2)$ and $E_{\sigma}(1, 1) = 1$, so the same limit gives (3.6). $\square$

The proof of (3.6) is thus contained in the Gram-entry argument. In particular, no unproved passage from pointwise slow variation to a sum down to its smallest indices is needed.

4 Positive congruence and uniform spectral control

4.1 An N -independent diagonal majorant

Proposition 4.1. *For every $0 < \eta < \rho$, put*

$$D_{\eta} = \text{diag}\{m^{-\eta/2} : m \geq 1\}. \quad (4.1)$$

There are positive finite-rank operators B_N and a constant $M = M(\sigma, L, \eta)$ such that

$$A_N = D_{\eta} B_N D_{\eta}, \quad \sup_N \|B_N\| \leq M. \quad (4.2)$$

In particular,

$$0 \leq A_N \leq M D_{\eta}^2, \quad 0 \leq E_{\sigma} \leq M D_{\eta}^2. \quad (4.3)$$

Proof. Choose $\varepsilon > 0$ so that $\eta + 2\varepsilon < \rho$. Define the finite matrix, extended by zero,

$$B_N(m, n) = (mn)^{\eta/2} A_N(m, n). \quad (4.4)$$

On the first N coordinates this is a congruence of A_N by the positive diagonal matrix with entries $m^{\eta/2}$. Thus B_N is positive. This definition is finite-dimensional and does not invoke an unbounded inverse of D_{η} on $\ell^2(\mathbb{N})$.

Set

$$s = \sigma + \varepsilon + \eta/2 < 1/2. \quad (4.5)$$

Proposition 3.2 gives $0 \leq B_N(m, n) \leq CE_s(m, n)$ for all indices, with C independent of N . If u, v are finitely supported complex sequences, write $|u| = (|u_m|)_m$ and similarly for v . Nonnegativity of the entries implies

$$\begin{aligned} |\langle u, B_N v \rangle| &\leq C \langle |u|, E_s |v| \rangle \\ &\leq C \|E_s\| \|u\| \|v\|. \end{aligned} \quad (4.6)$$

The boundedness of E_s follows from Theorem 2.1. Density therefore gives $\|B_N\| \leq M := C\|E_s\|$ uniformly in N . Direct comparison of matrix entries proves $A_N = D_\eta B_N D_\eta$. Since B_N is positive and bounded by MI , this factorization yields $0 \leq A_N \leq MD_\eta^2$.

For any finitely supported u , Proposition 3.3 permits passage to the limit in its finite quadratic form:

$$0 \leq \langle u, E_\sigma u \rangle = \lim_{N \rightarrow \infty} \langle u, A_N u \rangle \leq M \langle u, D_\eta^2 u \rangle.$$

Both E_σ and D_η^2 are bounded operators, so density extends the inequality to every $u \in \ell^2(\mathbb{N})$. This proves the second majorant in (4.3). $\square$

Corollary 4.2 (Uniform eigenvalue tails). *For every $0 < \eta < \rho$, the bounds (1.3) hold.*

Proof. The positive compact operator D_η^2 has ordered eigenvalues $j^{-\eta}$. Apply the min-max principle to (4.3). $\square$

The route from entries to spectrum in this argument is important. The entrywise estimate is used in the bilinear-form bound (4.6); it does not directly order the two positive matrices. The uniform norm bound on the positive congruence then produces the genuine operator inequality (4.3).

4.2 Finite heads and compact tails

Proposition 4.3. *The normalized Gram operators converge in operator norm:*

$$\|A_N - E_\sigma\| \rightarrow 0. \quad (4.7)$$

Proof. Fix any $\eta \in (0, \rho)$ and take B_N, M from Proposition 4.1. Let P_K project onto the first K coordinates and set $Q_K = I - P_K$. Since $\|D_\eta\| = 1$ and $\|Q_K D_\eta\| = (K+1)^{-\eta/2}$, the factorization gives

$$\|Q_K A_N\| \leq M(K+1)^{-\eta/2}. \quad (4.8)$$

The same estimate holds for $A_N Q_K$ by self-adjointness. Using $A_N - P_K A_N P_K = Q_K A_N + P_K A_N Q_K$, we obtain

$$\|A_N - P_K A_N P_K\| \leq 2M(K+1)^{-\eta/2}, \quad (4.9)$$

uniformly in N .

The operator E_σ is compact by Theorem 2.1, and P_K converges strongly to the identity. Hence

$$\|E_\sigma - P_K E_\sigma P_K\| \rightarrow 0 \quad (K \rightarrow \infty). \quad (4.10)$$

One may see this directly by first approximating E_σ in operator norm by a finite-rank operator; strong convergence of the projections is uniform on the finitely many vectors in a finite-rank representation, and then the approximation error is made arbitrarily small.

For each fixed K , entrywise convergence on the K by K head implies $\|P_K(A_N - E_\sigma)P_K\| \rightarrow 0$. The triangle inequality with (4.9) gives

$$\limsup_{N \rightarrow \infty} \|A_N - E_\sigma\| \leq 2M(K+1)^{-\eta/2} + \|E_\sigma - P_K E_\sigma P_K\|.$$

Let $K \rightarrow \infty$ to conclude (4.7). $\square$

Thus operator-norm convergence is supplied by a uniform tail estimate and finite-dimensional convergence. Entrywise convergence alone is not being used as a substitute for this argument.

5 The full ideal range and its sharpness

Proof of Theorem 1.1. The first two assertions are Corollary 4.2 and Proposition 4.3. It remains to prove the ideal convergence and its exact range.

Fix a real $q > 0$ with $q\rho > 1$. Choose

$$\frac{1}{q} < \eta < \rho, \quad (5.1)$$

and let M be a common constant for the two bounds in Corollary 4.2. Put $R_N = A_N - E_\sigma$. By Lemma 2.2 and positivity of A_N, E_σ ,

$$s_{2j-1}(R_N) \leq s_j(A_N) + s_j(E_\sigma) \leq 2Mj^{-\eta} \quad (j \geq 1). \quad (5.2)$$

The even singular value $s_{2j}(R_N)$ is no larger than the preceding one. Consequently,

$$\sum_{k \geq 1} s_k(R_N)^q \leq 2(2M)^q \sum_{j \geq 1} j^{-\eta q} < \infty, \quad (5.3)$$

uniformly in N . In particular, every R_N belongs to $\mathcal{S}_q$ in this range. For each fixed k , $s_k(R_N) \leq \|R_N\| \rightarrow 0$ by Proposition 4.3. The majorant in (5.2), grouped in pairs of indices, is q -summable because $\eta q > 1$. Dominated convergence for counting measure on $\mathbb{N}$ now gives

$$\|A_N - E_\sigma\|_{\mathcal{S}_q}^q = \sum_{k \geq 1} s_k(R_N)^q \rightarrow 0.$$

This proof uses scalar summation of singular values; no Banach-norm interpolation or triangle inequality is assumed when $q < 1$.

Finally suppose $q\rho \leq 1$. By Theorem 2.1, $E_\sigma \notin \mathcal{S}_q$. For every finite N , however, A_N is finite rank and hence belongs to $\mathcal{S}_q$. If $A_N - E_\sigma$ belonged to $\mathcal{S}_q$, closure under differences from Lemma 2.2 would imply $E_\sigma = A_N - (A_N - E_\sigma) \in \mathcal{S}_q$, a contradiction. This proves the stated nonmembership for every N , and completes the theorem. $\square$

The negative conclusion is stronger than failure of a uniform bound or failure of a chosen convergence argument. No finite Gram matrix can remove the non-summable tail of the classical limiting operator. Conversely, the positive proof requires no endpoint estimate for the approximating eigenvalues: the choice (5.1) uses only an exponent strictly between the ideal threshold and ρ .

6 Normalization, singular values, and admissible examples

6.1 Normalization by cumulative coefficient energy

The natural cumulative normalization is

$$F(N) = \sum_{k \leq N} |a(k)|^2, \quad C_N = F(N)^{-1} T_N^* T_N. \quad (6.1)$$

Corollary 6.1. *Under the standing assumptions, $\|C_N - E_\sigma\| \rightarrow 0$. For every $q > 0$ with $q\rho > 1$,*

$$\|C_N - E_\sigma\|_{\mathcal{S}_q} \rightarrow 0. \quad (6.2)$$

For $q\rho \leq 1$, $C_N - E_\sigma \notin \mathcal{S}_q$ for every N .

Proof. Proposition 3.3 gives

$$C_N = c_N A_N, \quad c_N = \frac{N^\rho L(N)^2}{\rho F(N)} \longrightarrow 1. \quad (6.3)$$

The scalars c_N are positive and bounded: they converge for large N , and only finitely many initial values remain. Theorem 1.1 also gives uniform boundedness of $\|A_N\|$. Thus

$$\|C_N - E_\sigma\| \leq |c_N - 1| \|A_N\| + \|A_N - E_\sigma\| \longrightarrow 0.$$

For every $0 < \eta < \rho$, the eigenvalues of C_N satisfy the same type of uniform bound $Cj^{-\eta}$, by (6.3) and Corollary 4.2. If $q\rho > 1$, choose η as in (5.1) and repeat the singular-value dominated-convergence argument of Section 5. This proves (6.2) also when $q < 1$. The negative statement uses finite rank and Theorem 2.1, exactly as before. $\square$

6.2 Ordered singular values and spectral moments

Corollary 6.2. *For every fixed $j \geq 1$,*

$$\frac{s_j(T_N)^2}{F(N)} \longrightarrow \lambda_j(E_\sigma) > 0. \quad (6.4)$$

After zero extension, the errors even satisfy

$$\sup_{j \geq 1} \left| \frac{s_j(T_N)^2}{F(N)} - \lambda_j(E_\sigma) \right| \longrightarrow 0. \quad (6.5)$$

For every real $q > 0$ with $q\rho > 1$,

$$\sum_{j \geq 1} \left(\frac{s_j(T_N)^2}{F(N)} \right)^q \longrightarrow \sum_{j \geq 1} \lambda_j(E_\sigma)^q. \quad (6.6)$$

Proof. The eigenvalues of C_N are precisely $s_j(T_N)^2/F(N)$. Apply (2.9) and Corollary 6.1 to obtain (6.5), hence (6.4). The strict positivity of every fixed limiting eigenvalue follows from positivity, compactness, and injectivity of E_σ on the infinite-dimensional space $\ell^2(\mathbb{N})$, as stated in Theorem 2.1.

For the last assertion, choose $\eta \in (1/q, \rho)$. The proof of Corollary 6.1 gives $\lambda_j(C_N) \leq Cj^{-\eta}$ uniformly in N, j , and Corollary 4.2 supplies the corresponding bound for E_σ . Pointwise convergence of the ordered eigenvalues and dominated convergence with the summable sequence $C^q j^{-\eta q}$ prove (6.6). $\square$

Equation (6.5) is an absolute-error statement. It does not assert a relative asymptotic uniformly over indices growing with N . In particular, the finite matrix has zero eigenvalues beyond its rank, whereas every fixed eigenvalue of E_σ is positive. For fixed j , combining (6.4) with (3.6) does give

$$s_j(T_N)^2 \sim \frac{\lambda_j(E_\sigma)}{\rho} N^\rho L(N)^2. \quad (6.7)$$

No joint j, N limit or new formula for the known LCM spectral constant is needed for this conclusion.

6.3 Nonmultiplicative and oscillatory examples

For every $\beta \in \mathbb{R}$, the function

$$L_\beta(x) = (1 + \log x)^\beta, \quad x \geq 1, \quad (6.8)$$

is positive and continuous, with positive lower and finite upper bounds on compact intervals. For fixed $t > 0$,

$$\frac{L_\beta(tx)}{L_\beta(x)} = \left(1 + \frac{\log t}{1 + \log x}\right)^\beta \rightarrow 1.$$

For $\beta = 1$, it is already nonmultiplicative on the integers:

$$L_1(2)L_1(3) = 1 + \log 6 + \log 2 \log 3 \neq 1 + \log 6 = L_1(6).$$

Since the power factor $k^{-\sigma}$ is completely multiplicative, the same comparison shows that $a(k) = k^{-\sigma}L_1(k)$ is not multiplicative even at the coprime pair 2, 3.

A bounded example with no limiting value is

$$L_{\text{osc}}(x) = 2 + \sin(\log \log(e + x)). \quad (6.9)$$

It lies between one and three. For each fixed $t > 0$,

$$\log \log(e + tx) - \log \log(e + x) = \log \left(\frac{\log(e + tx)}{\log(e + x)} \right) \rightarrow 0.$$

The sine function is Lipschitz and the denominator $L_{\text{osc}}(x)$ is at least one, so $L_{\text{osc}}(tx)/L_{\text{osc}}(x) \rightarrow 1$. The inner phase is continuous, increasing, and unbounded, hence there are sequences tending to infinity on which the sine equals 1 and -1 , respectively. Thus the slowly varying factor need not converge. Both examples satisfy all hypotheses of Theorem 1.1; an eventual constant is not being substituted for slow variation.

7 Scope of the result

The conclusion concerns one normalized Gram family with truncation parameter N . Its arithmetic input is the divisibility relation and the already known LCM operator. The extension established here is the allowance of arbitrary admissible pointwise slowly varying factors without multiplicativity, together with convergence throughout the exact positive-exponent ideal range. The uniform spectral tails and the consequences of Section 6 are components of that same result.

Several boundaries are worth stating explicitly. First, the hypotheses do not merely require regular variation of $\sum_{k \leq N} |a(k)|^2$ for an arbitrary arithmetic sequence. The pointwise representation $a(k) = k^{-\sigma}L(k)$ is what controls the two rescaled factors in each off-diagonal Gram entry. The more general multiplicative settings of [2] therefore remain distinct from the present theorem. Second, no convergence rate uniform over all admissible slowly varying functions is asserted. Third, the N -independent eigenvalue bound is proved for every $\eta < \rho$, not at its endpoint. Fourth, no simultaneous large-index relative singular-value asymptotic is claimed.

Finally, the LCM eigenvalue asymptotics, their spectral constant, and the prime-based factorization used to obtain them belong to [3]. The comparison with its even-integer Gram statement is bound to the accessed 2021 preprint version. The source comparison here does not certify global priority or an open problem in the later literature. No target Euler factors, functional equation, prescribed zero set, or Hilbert–Pólya realization is identified by this Gram-limit theorem.

Draft and reproducibility declarations

This is a mathematical research draft. The results are analytic and use no new numerical experiments, personal data, human participants, or animal studies. The accompanying source materials include the modular $\text{\LaTeX}$ manuscript, bibliography-verification record, and compilation receipt. AI assistance was used for mathematical transcription, literature cross-checking, and $\text{\LaTeX}$ preparation; this does not constitute external peer review. Author identities, affiliations, contributor roles, funding information, and competing interest declarations have not been supplied and must be completed and verified by the responsible authors before any journal submission.

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